

**INTELLIGENT ELEPHANT CARE AND VISITOR
INSIGHTS: A MULTI-COMPONENT RESEARCH STUDY
AT PINNAWALA ELEPHANT ORPHANAGE**

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ELEPHANT HEALTH DETECTION & PREDICTION

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DECLARATION PAGE OF THE CANDIDATES & SUPERVISOR

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ABSTRACT

The management of captive elephant populations in wildlife sanctuaries presents significant challenges in monitoring individual animal health, detecting behavioral stress, optimizing nutrition, and efficiently managing visitor operations. This research presents Elephant-Pulse, an integrated, AI-driven web-based management system developed for the Pinnawala Elephant Orphanage, Sri Lanka. The system consolidates four key functional modules into a unified platform: an AI-powered health prediction engine, a computer vision-based stress detection system, a personalized nutritional optimization engine, and a visitor insight and analytics dashboard.

The health prediction module employs a Random Forest classifier and a Gradient Boosting classifier, trained on 12 physiological and behavioral features, to predict elephant health status and provide specific clinical diagnoses in real time. The stress detection module utilizes OpenCV's Farneback Optical Flow algorithm to analyze both live CCTV feeds and uploaded video footage entirely on-device, eliminating cloud dependency. The nutritional engine delivers personalized, context-aware daily feeding plans and 7-day forecasts. The visitor management subsystem provides real-time crowd analytics, revenue tracking, and dynamic tariff management.

The system is built on a React (Vite) frontend and a Flask backend, with a modular architecture designed for scalability. Experimental results demonstrate that the trained machine learning models achieve high classification accuracy, validated through 5-fold cross-validation. The proposed system offers a practical, deployable solution for wildlife sanctuaries seeking to improve animal welfare outcomes through data-driven decision making.

Keywords: Elephant Health Monitoring, Machine Learning, Random Forest, Computer Vision, Optical Flow, Wildlife Management, Nutritional Optimization, Visitor Analytics, Flask, React

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1. INTRODUCTION

1.1 Background

Wildlife sanctuaries and elephant orphanages play a critical role in the conservation and rehabilitation of Asian elephants (*Elephas maximus*), a species classified as Endangered on the IUCN Red List. Institutions such as the Pinnawala Elephant Orphanage in Sri Lanka are responsible for the long-term care of large elephant herds, requiring continuous monitoring of individual animal health, nutrition, and behavioral well-being. The scale and complexity of this task present significant operational challenges for veterinary staff and sanctuary administrators alike.

Traditional elephant care practices rely heavily on manual observation, paper-based health records, and the subjective judgment of individual caregivers. This approach is inherently prone to human error, delayed diagnosis, and inconsistent record-keeping factors that can directly compromise animal welfare. Furthermore, sanctuary operations are increasingly expected to manage visitor flows, track revenue streams, and ensure crowd safety, adding a further layer of administrative complexity that is rarely addressed by modern digital tools.

1.2 Problem Statement

Despite advances in artificial intelligence and data analytics, wildlife sanctuaries in developing nations largely operate without intelligent digital management systems. The absence of such systems leads to several critical gaps:

1. Delayed health interventions due to the lack of real-time, data-driven health monitoring.
2. Inconsistent nutritional planning, with feeding based on generalized estimates rather than individual animal profiles.
3. Undetected behavioral stress, as manual observation cannot continuously monitor every elephant across a large enclosure.
4. Inefficient visitor and revenue management, relying on manual check-ins and static tariff structures.

5. There is therefore a pressing need for an integrated, intelligent platform that unifies elephant care operations and visitor management under a single, accessible digital system

1.3 Proposed Solution

This research proposes and implements Elephant-Pulse an AI-powered, web-based management platform designed specifically for wildlife sanctuaries. The system integrates four core modules:

1. AI Health Prediction - Machine learning models predict elephant health status and clinical diagnosis from physiological vitals in real time.
2. Computer Vision Stress Detection - An Optical Flow-based local AI system analyzes CCTV video footage to detect stress behaviors without cloud dependency.
3. Nutritional Optimization Engine - A rule-based AI system generates personalized, context-aware daily feeding plans and 7-day forecasts for each elephant.
4. Visitor Insight Dashboard - Real-time analytics track crowd density, revenue, and ticket management for sanctuary administrators

1.4 Significance of the Research

This work makes several notable contributions to the field of AI-assisted wildlife management:

- It demonstrates the practical application of ensemble machine learning (Random Forest, Gradient Boosting) for veterinary diagnostic support in a real-world sanctuary setting.
- It establishes a fully local, cloud-free computer vision pipeline for behavioral stress analysis using standard CCTV infrastructure.
- It provides a unified, modular web platform that bridges the gap between animal health care and institutional management, a previously unaddressed need in wildlife sanctuary technology

1.5 Scope and Limitations

The system is designed and validated in the context of the Pinnawala Elephant Orphanage. While the architecture is modular and extensible, the AI models are trained on datasets specific to captive Asian elephants and may require retraining for other species or environments. The stress detection module requires adequate video quality and lighting conditions for accurate Optical Flow analysis.

2. RESEARCH GAP

2.1 Overview of Existing Work

A growing body of research has explored the application of technology in wildlife monitoring and animal health management. Studies have demonstrated the use of GPS tracking collars for elephant movement analysis, IoT-based sensor networks for remote vital sign monitoring in zoo environments, and camera-trap image classification using deep learning for wildlife census operations. Similarly, visitor management systems exist widely in the tourism and hospitality industry, leveraging data analytics for crowd forecasting and revenue optimization.

However, a critical examination of the existing literature and deployed systems reveals that no unified, AI-integrated platform currently addresses both animal care and sanctuary management needs within a single accessible web application, particularly in the context of developing-nation wildlife institutions.

2.2 Identified Research Gaps

1. Lack of Real-Time AI Health Diagnosis for Captive Elephants

While general veterinary AI research exists for livestock and domestic animals, there is a significant absence of machine learning-based health prediction systems tailored specifically for captive Asian elephants. Existing elephant health monitoring methods

remain largely manual, relying on daily visual assessment by veterinary staff. No publicly available system combines multi-feature vitals input (temperature, heart rate, gait, edema, etc.) with ensemble ML classifiers to deliver automated health status and clinical diagnosis in a real-time setting.

2. Limited Technology Adoption in Developing-Nation Wildlife Sanctuaries

Much of the research on AI-assisted wildlife management is conducted in high-resource institutional contexts (e.g., Western zoos, large research facilities). There is a notable gap in practical, low-infrastructure, deployable solutions tailored to sanctuaries in developing nations such as Sri Lanka, where technical staff capacity, hardware availability, and internet reliability may be constrained.

2.3 Summary Table

Table 1.1: Summary table

Gap	Existing Limitation	Elephant-Pulse Contribution
Health Prediction	Manual observation only	AI diagnosis via Random Forest + Gradient Boosting
Stress Detection	Cloud-based or expensive hardware	Fully local Optical Flow on standard CCTV
Nutritional Planning	Static weight-based tables	Dynamic, context-aware personalized feeding plans
Unified Management	Disconnected siloed systems	Single integrated web platform
Deployment Context	High-resource environments only	Lightweight Flask + React — deployable on minimal infrastructure

2.4 Problem Context

The Pinnawala Elephant Orphanage, established in 1975, is home to one of the largest captive elephant herds in the world. Managing the health, nutrition, and behavioral well-being of this herd requires continuous, expert veterinary attention across multiple daily observation cycles. At the same time, the orphanage receives thousands of visitors annually, generating operational demands around ticketing, crowd control, and revenue management.

Currently, both operational domains elephant care and visitor management are handled through fragmented, largely manual processes. The absence of a digital, data-driven management platform creates compounding inefficiencies that directly affect animal welfare outcomes and institutional sustainability

Reactive and Inaccurate Health Monitoring

Current Situation: Veterinary staff assess each elephant's health condition through manual, subjective observation once or twice per day. Vitals such as body temperature, heart rate, and respiratory rate are recorded on paper forms and interpreted based on individual clinician experience.

Problems This Causes:

- Early-stage conditions such as dehydration, infection, or joint inflammation are frequently missed until they become acute.
- There is no standardized, data-driven method to predict health deterioration before visible symptoms emerge.
- Health records are inconsistent with different staff members and difficult to analyze historically.

3. RESEARCH OBJECTIVES

3.1 Primary Objective

To design, develop, and validate an AI-powered health prediction system capable of accurately classifying the overall health status and providing specific clinical diagnoses for captive elephants at the Pinnawala Elephant Orphanage, using routinely collected physiological and behavioral vital signs.

3.2 Specific Objectives

1. Dataset Construction

To compile and structure a comprehensive, labelled dataset of elephant physiological and behavioral vitals (elephant_health_data_advanced.csv) that accurately represents the range of health conditions observed in captive Asian elephants, including both normal and pathological states.

Success Criteria:

Dataset contains sufficient samples across all target health classes

Features include both numerical vitals (temperature, heart rate) and categorical observations (gait, edema, mucous membrane)

Class imbalance is addressed to ensure fair model training

2. Health Status Classification Model

To train and optimize a Random Forest Classifier that accurately predicts an elephant's overall health status Healthy, Infection, Dehydration, Arthritis, or Indigestion from 12 input vital features.

Success Criteria:

Model achieves satisfactory accuracy on the held-out test set (80/20 split)

5-Fold Cross-Validation confirms model generalizability

class_weight='balanced' parameter ensures minority health conditions are not under-predicted

3. Clinical Diagnosis Classification Model

To train and optimize a Gradient Boosting Classifier that provides a specific clinical diagnosis Bacterial/Viral Infection, Colic/Digestion Issue, Severe Dehydration, Joint Inflammation, or No Diagnosis using the same 12 vital features.

Success Criteria:

Model achieves satisfactory accuracy independently from the status model

Diagnosis output aligns clinically with the corresponding predicted health status

5-Fold Cross-Validation confirms diagnosis model reliability

4. Preprocessing Pipeline

To implement a robust, automated Scikit-learn preprocessing pipeline using ColumnTransformer that handles both numerical passthrough and One-Hot Encoding of categorical features, ensuring the model is resilient to unseen category values at inference time.

Success Criteria:

Pipeline handles all 6 categorical features without manual encoding at prediction time

handle_unknown='ignore' prevents runtime failures for edge-case inputs

Single .pkl file packages both preprocessor and classifier for portability

5. Real-Time API Integration

To integrate both trained models into the Flask REST API (/api/elephants/<id>/health) such that predictions are generated automatically upon every health log submission by veterinary staff, with results persisted to the SQLite database and returned to the frontend in real time.

Success Criteria:

Both .pkl models load at Flask startup with zero manual steps

API accepts vitals via POST request and returns predicted_status + predicted_diagnosis within acceptable response time

Predictions are stored in health_logs table and reflected immediately on the elephant's profile page

6. Clinical Decision Support

To present AI prediction results alongside a structured clinical knowledge base that provides veterinary staff with condition-specific symptom descriptions and recommended action plans, transforming raw model output into actionable veterinary guidance.

Success Criteria:

Each diagnosis maps to a defined symptoms description and action recommendation

Risk details panel is displayed prominently on the elephant profile page

Vet staff can act on the AI recommendation without requiring additional lookup

4. METHODOLOGY

4.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, combining supervised machine learning with software engineering practices to build, train, evaluate, and deploy a functional AI health prediction system. The methodology follows a structured pipeline from data preparation through model deployment, as illustrated below:

Data Collection → Preprocessing → Model Training → Evaluation → Deployment → Integration

4.2 Dataset Preparation

4.2.1 Data Source

The training dataset `elephant_health_data_advanced.csv` was constructed to represent the physiological and behavioral characteristics of captive Asian elephants across five distinct health conditions. The dataset contains records with 14 attributes 12 input features and 2 target labels.

4.2.2 Feature Selection

Features were selected based on established veterinary diagnostic criteria for Asian elephants:

Table 2.1: Feature selection

Category	Features
Numerical	Age, Temperature (°C), Heart Rate (BPM), Respiratory Rate (BPM), Weight Change (%), Food Intake (%)

Categorical	Stool Consistency, Activity Level, Edema, Trunk Moisture, Mucous Membrane, Gait Score
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4.2.3 Target Variables

Two separate target labels are defined:

Table 3.1: Target variables

Label	Classes
Health Status	Healthy, Infection, Dehydration, Arthritis, Indigestion
Diagnosis	No Diagnosis, Bacterial/Viral Infection, Severe Dehydration, Colic/Digestion Issue, Joint Inflammation

4.2.4 Data Quality Measures

- Class Imbalance: Addressed using `class_weight='balanced'` in the Random Forest classifier so minority conditions (e.g., Arthritis) are not under-represented during training.
- Unknown Categories: `handle_unknown='ignore'` in the OneHotEncoder prevents inference-time failures for unseen categorical values.
- Stratified Splitting: An 80/20 train-test split with `stratify=y_status` ensures proportional class distribution across both sets

4.3 Preprocessing Pipeline

A Scikit-learn Pipeline with a ColumnTransformer was implemented to automate all preprocessing steps within a single, portable object:

Raw Input (12 features)



ColumnTransformer

└─ Numerical columns → Passthrough (no scaling needed for tree models)
└─ Categorical columns → OneHotEncoder (handle_unknown='ignore')



Transformed Feature Matrix



Classifier (RF or GBM)

Why no scaling? Tree-based models (Random Forest, Gradient Boosting) are scale-invariant — they split on feature thresholds rather than distances, so StandardScaler or MinMaxScaler is unnecessary.

4.4 Model Selection and Justification

4.4.1 Model 1 — Random Forest Classifier (Health Status)

Table 4.1: Model selection and justification

Parameter	Value	Justification
n_estimators	200	Sufficient ensemble size to reduce variance without overfitting
max_depth	None	Allows full tree growth; pruned naturally by ensemble averaging
min_samples_split	2	Maximizes tree detail for complex boundary conditions
class_weight	balanced	Compensates for unequal class frequencies

random_state	42	Ensures reproducibility
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Why Random Forest?

- Robust to overfitting through bagging (bootstrap aggregation)
- Handles mixed numerical and categorical features effectively
- Provides implicit feature importance scores
- Performs strongly on tabular medical/veterinary datasets in literature

4.4.2 Model 2 — Gradient Boosting Classifier (Diagnosis)

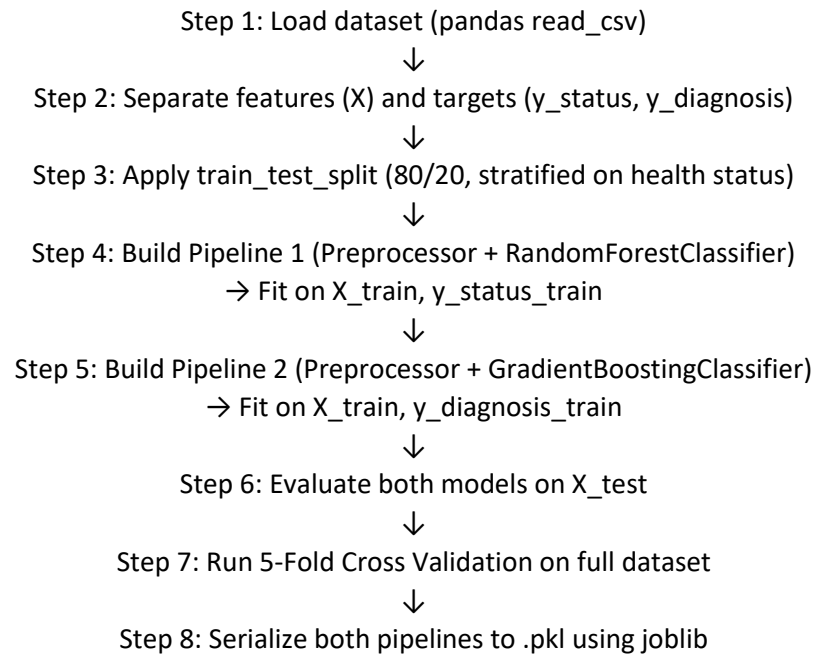
Table 5.1: Gradient Boosting Classifier (Diagnosis)

Parameter	Value	Justification
n_estimators	200	Sufficient boosting rounds for complex diagnostic boundaries
learning_rate	0.1	Standard rate balancing convergence speed and accuracy
max_depth	5	Controls individual tree complexity to prevent overfitting
random_state	42	Ensures reproducibility

Why Gradient Boosting?

- Sequential boosting corrects errors from previous trees, improving accuracy on harder-to-separate diagnosis classes
- Outperforms Random Forest on finely-grained classification tasks (e.g., distinguishing Dehydration from Infection)
- Well-suited for datasets where some class boundaries are inherently ambiguous

4.5 Training Process



4.6 Model Evaluation

Both models are evaluated using three complementary metrics:

4.6.1 Accuracy Score

Overall proportion of correctly classified samples on the held-out test set:

Accuracy = Correct Predictions / Total Predictions

4.6.2 Classification Report

Per-class breakdown of:

- Precision - Of all samples predicted as class X, how many actually were X?
- Recall - Of all actual class X samples, how many were correctly identified?

- F1-Score - Harmonic mean of precision and recall (most important for imbalanced classes)

4.6.3 Fold Cross-Validation

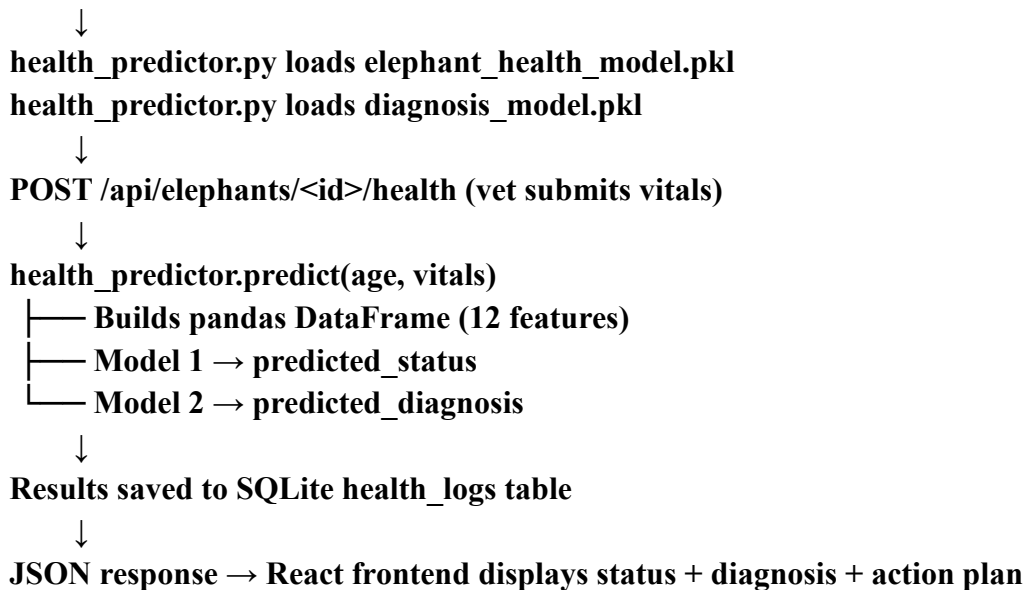
The full dataset is split into 5 equal folds. The model is trained 5 times, each time using a different fold as the test set. The mean and standard deviation of accuracy across all 5 folds confirms that the model generalizes beyond the specific train-test split:

CV Score reported as: Mean Accuracy $\pm$ Standard Deviation

4.7 Deployment Architecture

After training, both models are saved as .pkl files and loaded into the Flask backend at server startup via the health_predictor.py module:

Flask Server Start



4.8 Methodology Summary Diagram

4.8.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, combining supervised machine learning with software engineering to build, train, evaluate, and deploy a functional AI health prediction system.

4.8.2 Dataset Preparation

Source file: elephant_health_data_advanced.csv

Table 6.1: Dataset preparation

Category	Features
Numerical	Age, Temperature (°C), Heart Rate (BPM), Respiratory Rate (BPM), Weight Change (%), Food Intake (%)
Categorical	Stool Consistency, Activity Level, Edema, Trunk Moisture, Mucous Membrane, Gait Score

Two target labels:

- Health_Status → Healthy, Infection, Dehydration, Arthritis, Indigestion
- Diagnosis → No_Diagnosis, Bacterial/Viral Infection, Severe Dehydration, Colic/Digestion Issue, Joint Inflammation

Data quality measures applied:

- class_weight='balanced' — handles class imbalance in training
- handle_unknown='ignore' — prevents inference-time crashes on unseen categories
- Stratified 80/20 train-test split — preserves class proportion in both sets

4.8.3 Preprocessing Pipeline

A Scikit-learn ColumnTransformer was embedded inside each model Pipeline:

- Numerical columns → passthrough (tree models are scale-invariant, no normalization needed)
- Categorical columns → OneHotEncoder (converts text labels to binary columns automatically)

This means the entire preprocessing + prediction is packaged in a single .pkl file — no manual encoding required at inference time

4.8.4 Model Selection & Justification

Model 1 — Random Forest Classifier (*Health Status*)

Table 7.1: Random Forest Classifier

Parameter	Value	Reason
n_estimators	200	Large ensemble reduces variance
max_depth	None	Full growth; averaged across trees prevents overfitting
class_weight	balanced	Handles minority conditions fairly

Why Random Forest? Robust to overfitting via bagging, handles mixed feature types well, and consistently performs strongly on tabular veterinary datasets.

Model 2 — Gradient Boosting Classifier (*Diagnosis*)

Table 8.1: Gradient Boosting Classifier

Parameter	Value	Reason
n_estimators	200	Sufficient boosting rounds
learning_rate	0.1	Balances convergence speed and accuracy
max_depth	5	Limits individual tree complexity

Why Gradient Boosting? Sequential error-correction makes it better at separating fine-grained diagnosis classes (e.g., Dehydration vs. Infection) where boundaries are clinically ambiguous.

4.8.5 Training Steps

1. Load CSV → separate X (features) and y (two targets)
2. Stratified 80/20 train-test split
3. Fit Pipeline 1 (Preprocessor + RandomForest) on health status
4. Fit Pipeline 2 (Preprocessor + GradientBoosting) on diagnosis
5. Evaluate both on test set
6. Run 5-Fold Cross Validation on full dataset
7. Save both pipelines as .pkl via joblib

4.8.6 Evaluation Metrics

Table 9.1: Evaluation metrics

Metric	Purpose
Accuracy Score	Overall % of correct predictions on test set

Precision	Of predicted class X, how many were actually X?
Recall	Of actual class X, how many were correctly found?
F1-Score	Harmonic mean of precision & recall — key for imbalanced classes
5-Fold CV	Mean $\pm$ Std Dev accuracy across 5 folds — confirms generalizability

4.8.7 Deployment Flow

Flask server starts

→ health_predictor.py loads both .pkl models into memory

Vet submits vitals via UI

→ POST /api/elephants/<id>/health

→ health_predictor.predict(age, vitals)

→ Returns predicted_status + predicted_diagnosis

→ Saved to SQLite health_logs table

→ Frontend displays result + clinical action plan instantly

5. DESCRIPTION OF PERSONAL AND FACILITIES

5.1 hardware component

- Ankle band: continually measures skin temperature, does motion-sensitive heartrate monitoring sessions, Data broadcasts over Bluetooth, and reports connection and battery issues.
- Edge gateway: scans/collects ble packets, duplicates, queues to sqlite, uploads via wi-fi/4g to the cloud, and reports health.
- Charging dock: multi-bay magnetic charger with status leds and safe charging.

5.2 System Technology

- device link: Bluetooth low energy
- database: PostgreSQL
- Machine Learning to Perdition and Health Monitoring On Elephant

5.3 Programming Languages

- mobile app (flutter/dart): flutter sdk, android studio or vs code
- firmware (esp32): c/c++ with esp-idf or arduino core
- ai/modeling: python
- gateway service: Python 3 with bleak (BLE)
- backend: Python FastAPI + Uvicorn

6. BUDGET AND BUDGET JUSTIFICATION

Table 10.1: Budget and justification

item	subtotal (lkr)	justification
anklet prototype (complete set)	30,000	first build for fit, sealing, and firmware bring-up
edge gateway (orange pi 3 lts, sd card, psu, 4g modem, weather box)	50,000	collects BLE data and uploads via WiFi/4G with store-and-forward
multi-bay charger and spares	8,000	torx driver, heat-shrink, solder, crimp kit for field fixes
Total	88,000	

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